

# MetaSurfaceToolkit

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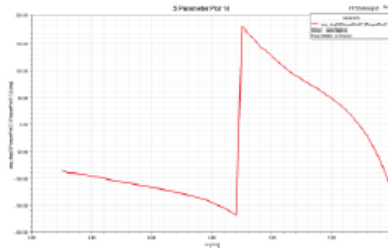
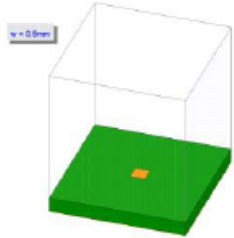
# Reflectarray or Transmitarray Design Procedure

## Unit cell Selection

- Subwavelength unit cells
- $|T| \approx 1$  or  $|r| \approx 1$
- Phase ( $\phi$ ) variation of  $360^\circ$



## Phase tuning of Unit cell



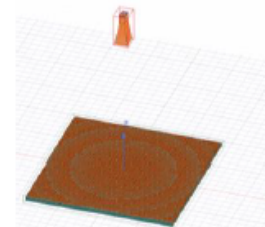
- S11 for Reflectarray
- S21 for Transmitarray

## Prototype

$$\Phi_{RA} = k_0 (D_n - \sin \theta_0 (x_n \cos \varphi_0 + y_n \sin \varphi_0)) \quad (1) \text{ (Reflectarray phase distribution)}$$

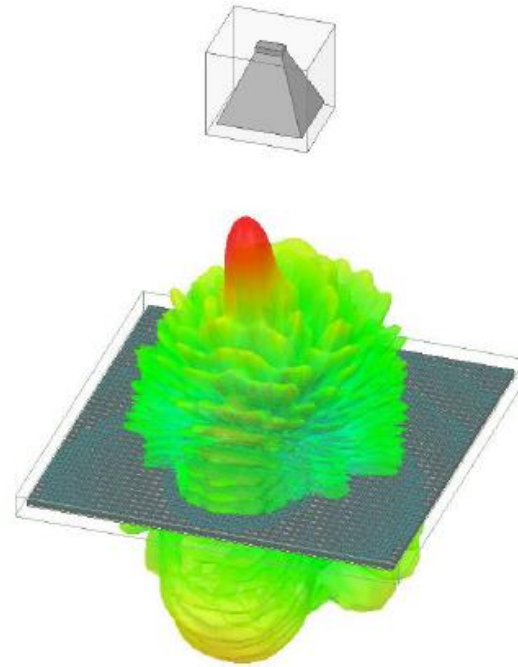
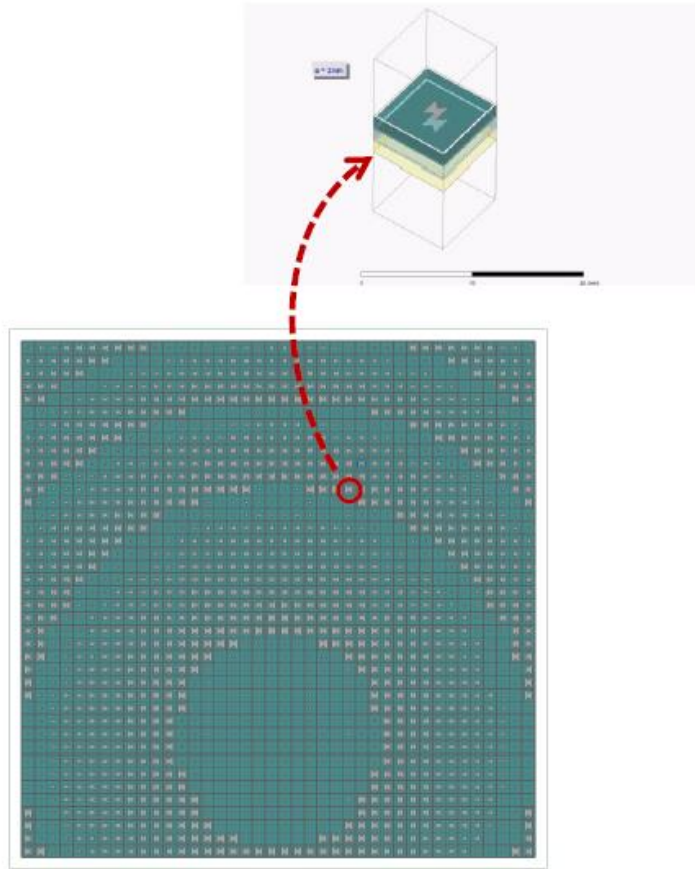
$$\Phi_{TA} = k_0 D_n \quad (2) \text{ (Transmitarray phase distribution)}$$

$$D_n = \sqrt{((x_n - x_f)^2 + (y_n - y_f)^2 + (z_f)^2)} \quad (3)$$

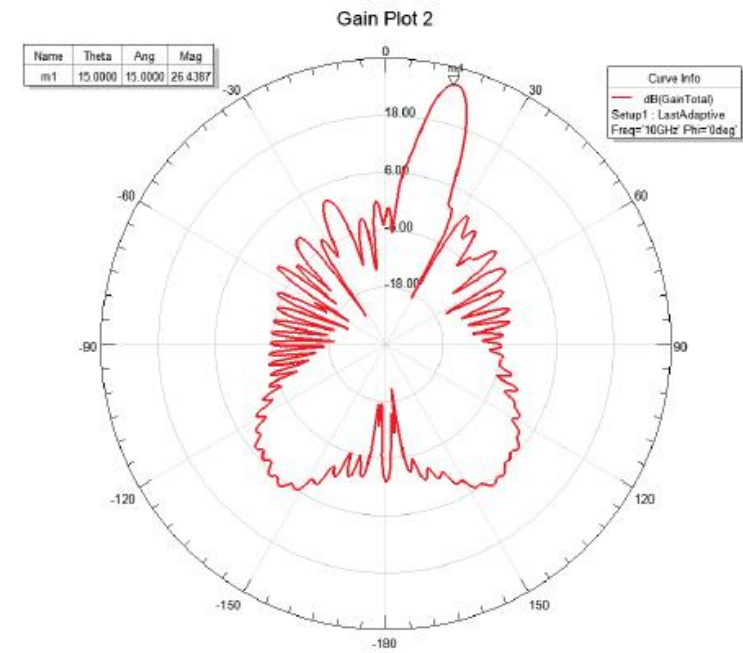


- $(x_f, y_f, z_f)$  is the phase center of the feed antenna
- $(x_n, y_n)$  is the position coordinates of the  $n^{\text{th}}$  element
- $(\vartheta_0, \varphi_0)$  is the beam collimation direction

# Case1: Reflectarray at 10GHz

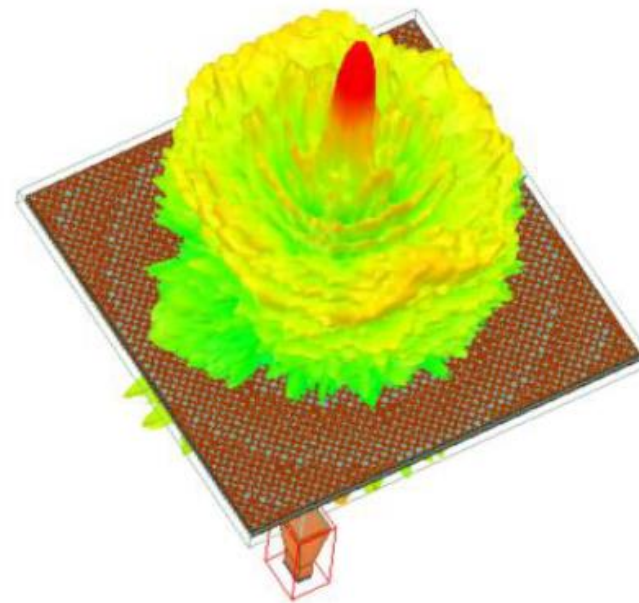
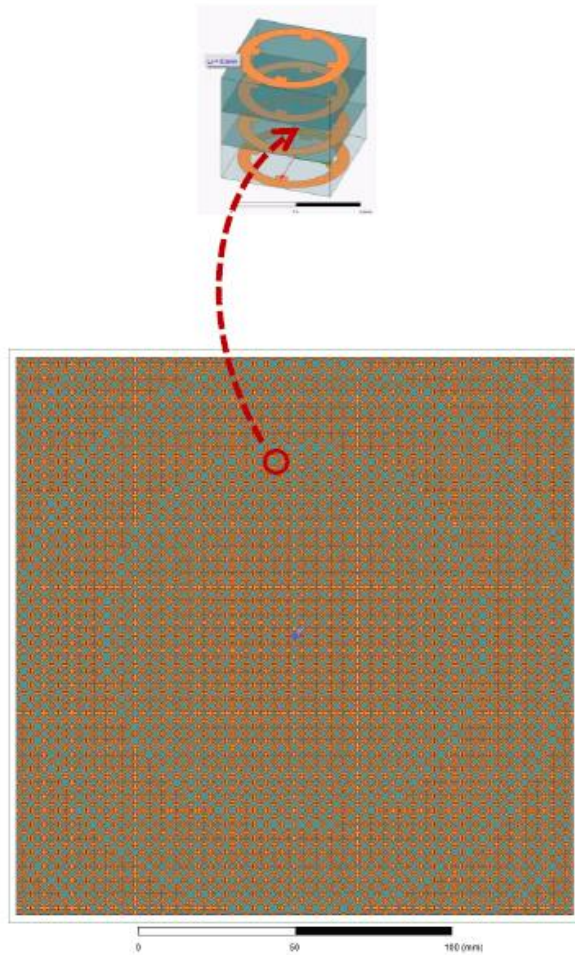


3D Radiation Pattern



2D Radiation Pattern

## Case2: Transmitarrayat 28GHz



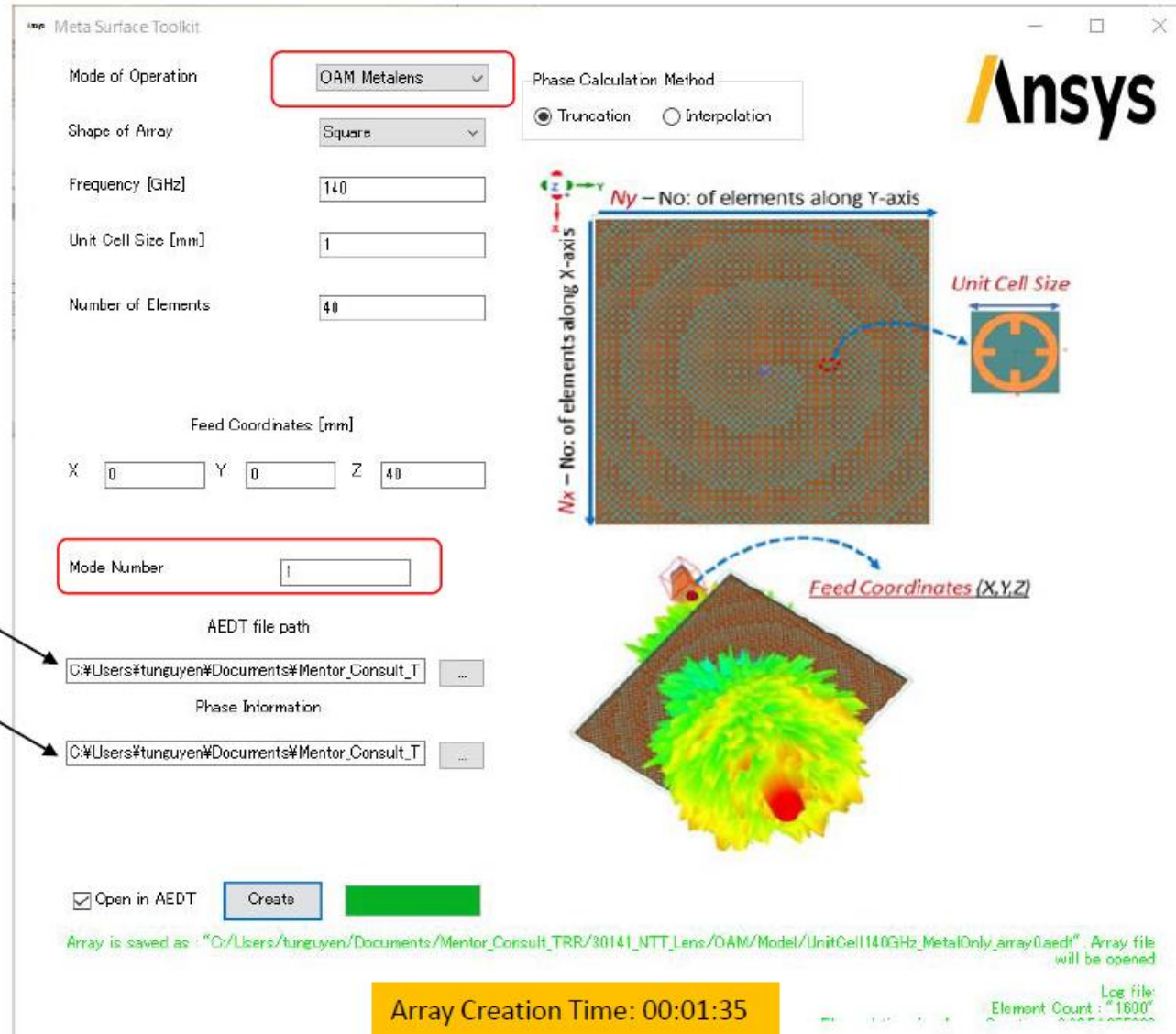
3D Radiation Pattern



2D Radiation Pattern

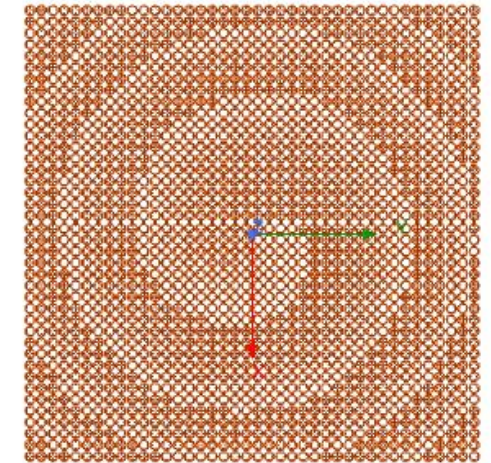


# OAM Meta Lens with Meta Surface Toolkit

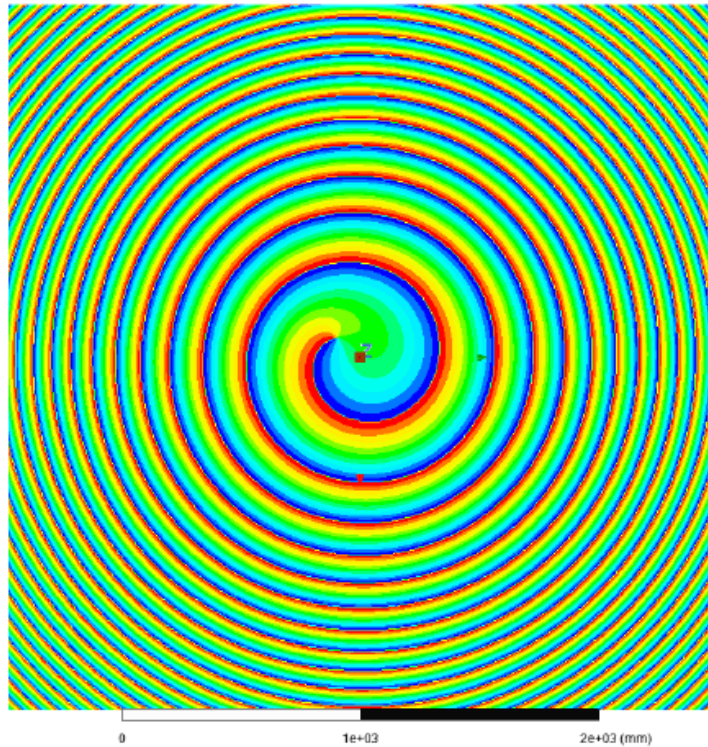


New Toolkit:

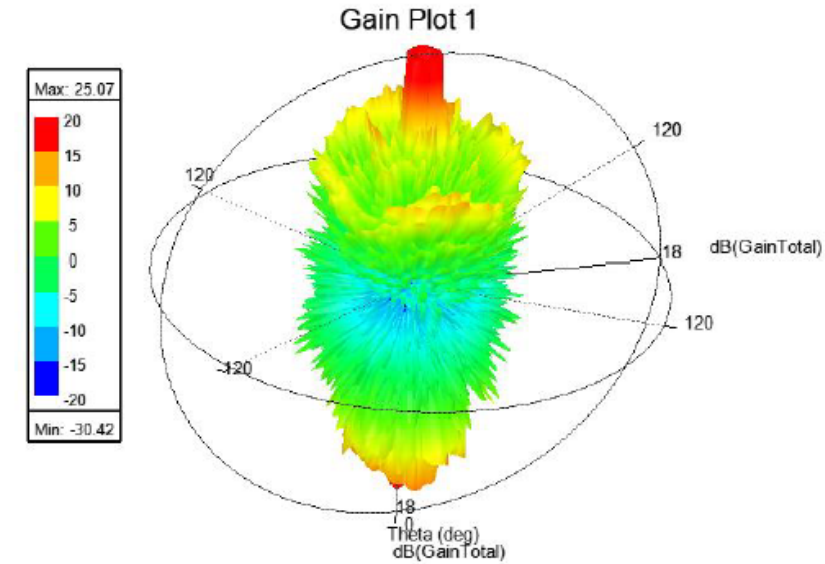
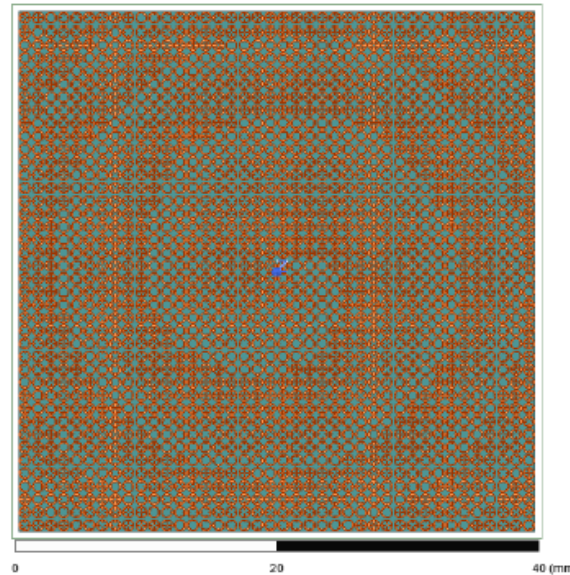
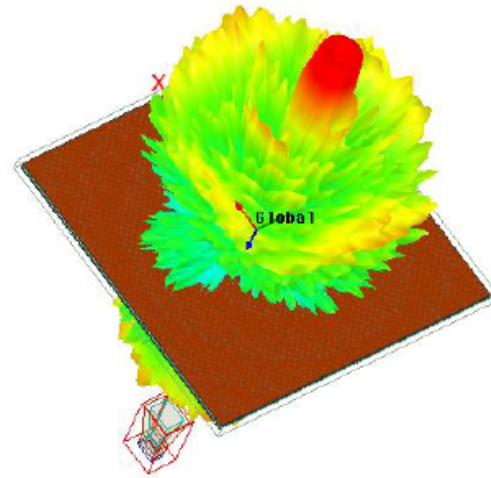
Add OAM Metalens design mode. User does not need to provide the profile information (phase distribution vs location), the Toolkit will automatically generate this information based on the feed coordinate and specified OAM mode.



# OAM Design for 140GHz (Mode 1) -Results



Phase Profile – Mode1

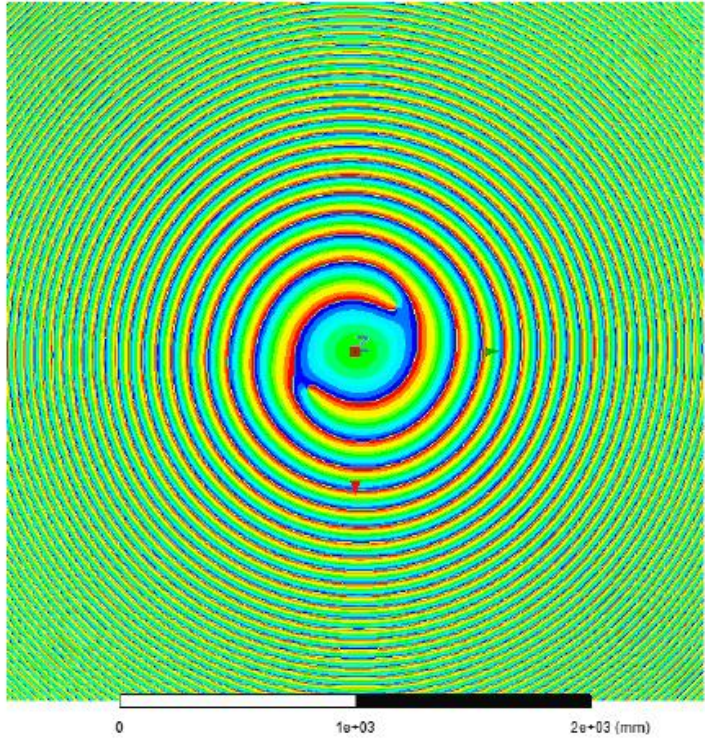


Radiation Pattern OAM Mode 1

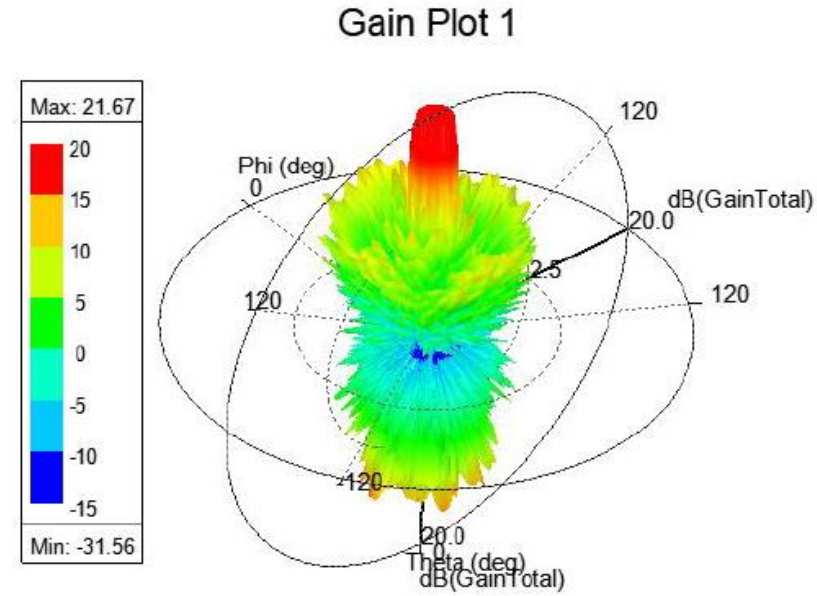
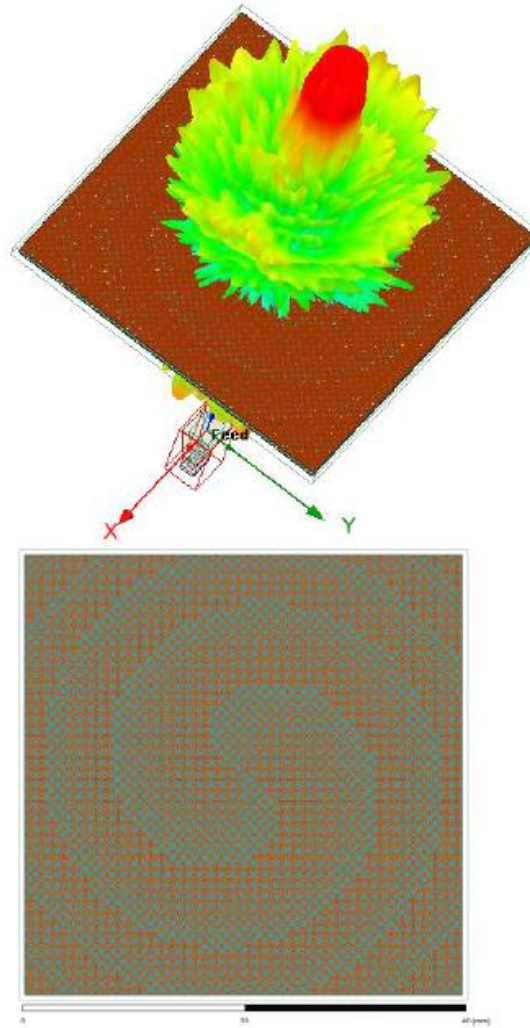
Simulation Time: 12:16:23  
Memory Usage: 450 GB  
No. of Cores: 40



# OAM Design for 140GHz (Mode 2)



Phase Profile – Mode2



Radiation Pattern OAM Mode 2

Simulation Time: 20:26:26  
Memory Usage: 612 GB  
No. of Cores: 40